

Let $I = (a, f) \subseteq R[x]$ be an ideal generated by $a, f \in R[x]$.

Consider the diagram:

$$\begin{array}{ccccc} R[x] & \xrightarrow{\phi_a} & (R/aR)[x] & \xrightarrow{\pi_f} & (R/aR)[x]/(\overline{f}) \\ & \searrow \pi_a & \nearrow \cong & & \\ & R[x]/aR[x] & & & \end{array}$$

$$\begin{array}{ll} R[x] \rightarrow R[x]/aR[x] & R[x]/aR[x] \rightarrow (R/aR)[x] \\ p(x) \mapsto p(x) + aR[x] & p(x) + aR[x] \mapsto \overline{p(x)}, \text{ where } \overline{\sum a_n \cdot x^n} = \sum (a_n + aR) \cdot x^n \\ \text{onto Homomorphism} & \text{Isomorphism.} \end{array}$$

$\Rightarrow \phi: R[x] \rightarrow (R/aR)[x]$ is onto Homomorphism, so is $\pi_f \circ \phi$ onto Homomorphism. Moreover,

$$\begin{aligned} \ker(\pi_f \circ \phi) &= (\pi_f \circ \phi)^{-1}[\{0\}] = \phi^{-1}[\ker \pi_f] = \phi^{-1}[(\overline{f})] \\ &= \{p(x) \in R[x] \mid \overline{p(x)} \in \ker \pi_f\} = \{p(x) \in R[x] \mid \overline{p(x)} \in (\overline{f(x)})\} \\ (*) &= \{p(x) \in aR[x] + f(x)R[x]\} = (a) + (f) = (a, f) = I. \end{aligned}$$

Assume $a \in R$ and $(a) \subseteq R$ is maximal, and $\overline{f} \in (R/aR)[x]$ is irreducible.

Then, R/aR is a field, and therefore $(R/aR)[x]$ is p.i.d.

So, $(\overline{f})$ is maximal, it implies that $(R/aR)[x]/(\overline{f})$ is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x] + fR[x]) \cong (R/aR)[x]/\overline{f} \cdot (R/aR)[x]$$

Note: (*) is obtained by: given $p(x) \in R[x]$,

$$\overline{p(x)} \in (\overline{f(x)}) \Leftrightarrow \overline{p(x)} = \overline{g(x)} \cdot \overline{f(x)} \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) - g(x)f(x) \in aR[x] \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) \in (a) + (f(x)) = (a, f).$$

Investigate $\mathbb{Z}[x]/(6, x^2+x+1)$

1). $(6, x^2+x+1)$

1-1). x^2+x+1 is irreducible.

x^2+x+1 reducible $\Rightarrow x^2+x+1 = (x+a)(x+b)$ s.t. $ab=1$

So $ab=1 \Rightarrow a=b=1$ or $a=b=-1$, but ± 1 are not roots,

1-2). $(6, x^2+x+1) = (2, x^2+x+1) \cap (3, x^2+x+1)$

Denote $f = x^2+x+1$. Then,

$$(6, f) = (6) + (f) = (2)(3) + (f) = [(2) + (f)] \cdot [(3) + (f)] = (2, f) \cdot (3, f) = (2, f) \cap (3, f)$$

Thus, by CRT,

$$\mathbb{Z}[x]/(6, f) \cong (\mathbb{Z}[x]/(2, f)) \times (\mathbb{Z}[x]/(3, f))$$

2). Consider

$$\mathbb{Z}[x] \rightarrow (\mathbb{Z}/(2))[x] \rightarrow (\mathbb{Z}/(2))[x]/(\bar{f})$$

$$\sum a_n x^n \mapsto \sum (a_n \bmod 2) x^n \mapsto \sum (a_n \bmod 2) x^n + (\bar{f})$$

which is onto Homomorphism with kernel $(6, f)$.

Since $\bar{f} = \bar{x}^2 + \bar{x} + 1$ has no root in $\mathbb{F}_2 \cong \mathbb{Z}/(2)$, so irreducible, thus

$$\mathbb{Z}[x]/(2, f) \cong \mathbb{F}_2[x]/(\bar{x}^2 + \bar{x} + 1) \cong \mathbb{F}_{2^2}.$$

But, $\bar{f}$ is not irreducible over $\mathbb{Z}/(3) \cong \mathbb{F}_3$, as $\bar{x}^2 + \bar{x} + 1 = (\bar{x} + 2)^2$, so

$$\mathbb{Z}[x]/(3, f) \cong \mathbb{F}_3[x]/(\bar{x}^2 + \bar{x} + 1)$$

By the division algorithm, given $f(x) \in \mathbb{Z}[x]$,

$$f(x) = b(x) \cdot (x^2+x+1) + ax + b \text{ for some } b(x) \in \mathbb{Z}[x] \text{ and } a, b \in \mathbb{Z},$$

$$\text{So that } \mathbb{F}_3[x]/(\bar{x}^2 + \bar{x} + 1) = \{ax + b + (\bar{x}^2 + \bar{x} + 1) \mid a, b \in \mathbb{F}_3\},$$

$$(a\bar{x} + b) \cdot (c\bar{x} + d) = ac\bar{x}^2 + (ad + bc)\bar{x} + bd = (2ac + ad + bc)\bar{x} + ac + bd$$

If $a=0$, then b is an element of a field $\mathbb{F}_3$, so it is a unit unless it is zero.

$$\text{If } a=1, \text{ then } (2c + d + bc)\bar{x} + c + bd = 1$$